

Bus Controller Design Specification

Ver0.4

DM270

Department: IMAGING

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HISTORY

Version	Date	Author	Notes
Ver 0.0	02/28/02	Toshiki Suzuki	Initial Release for DM270
Ver 0.1	03/25/02	Toshiki Suzuki	Spec change, added I2C
Ver 0.2	05/09/02	Toshiki Suzuki	Top I/O port change for CPRM CPRM ROM size change Added MG module
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Ver 0.4	06/17/02	Toshiki Suzuki	Reflect changes after spec review. (added uart*_nwait, changed port names)

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1. ABSTRACT

All RAM, ROM and memory-mapped registers can be accessed by ARM core through Bus Controller (BUSE). BUSE is designed to control the address and the data, which are requested by ARM for Digital Media chip. BUSE has capability of accessing up to 0xFFFFFFFF of address.

2. REFERENCES

ARM7TDMI datasheet ARM DDI 0029E, Aug. 1996, ARM Ltd.

3. GLOSSARY

AIM	ARM Internal Memory
ARM	ARM7TDMIE
BUSE	Bus Controller
HPIB	Host Peripheral Interface Bridge
MTC	Memory Traffic Controller
RVR	Reset Vector ROM
USB	Universal Serial BUS

4. FUNCTION

BUSE contains sub-blocks called adrdecode, which generates chip select signal to activate each peripheral block. The adrctl block controls the address, the data and control signals among the peripheral modules. The waitctl controls the number of wait-state to adjust read/write timing with being different speed peripheral block. The register of BUSE are implemented in reg block, which includes revr block which is the lower hierarchy block in reg block in order to be easy modification on netlist level.

The design target for BUSE module is as follows:

- ◆ Full addressing from 0x0 to 0xFFFFFFFF can be done.
- ◆ The number of wait-state are inserted automatically according to the peripheral module accessed by ARM.
- ◆ Internal RAM which contains RTOS and the application program of ARM can be accessed by 32 bits data bus.
- ◆ All registers of peripheral module are accessed with 16 bits data bus in no wait accessing except USB, SD/MMC IF, MS and MG module.
- ◆ Wait-state is inserted with a wait signal from the peripheral module which is synchronized on clk_arm clock during memory access or some register accesses.
- ◆ Generate chip select signal to select memory-mapped area.
- ◆ Support the converting endian of data.
- ◆ The alignment fault is not support on addressing.
- ◆ Support three type of data access mode, e.g. a byte, a half-word and a word which is 8 bits, 16 bits and 32 bits respectively during data or instruction access to the external memories like SDRAM, Flash Memory and so on.
- ◆ Outputs CS for external memory area. The partition of the external memory area is controlled by Memory Traffic Controller Module. It's suitable for various combination of external memory.
- ◆ Generate data strobe signals to access memory through other memory interface module.
- ◆ To reduce latency and power consumption while accessing program memory, memory bus is separated from peripheral bus.

4.1 Block Diagram

The following block diagram shows the inter-connect method used for ADRDECODE, WAITCTL and some logic.

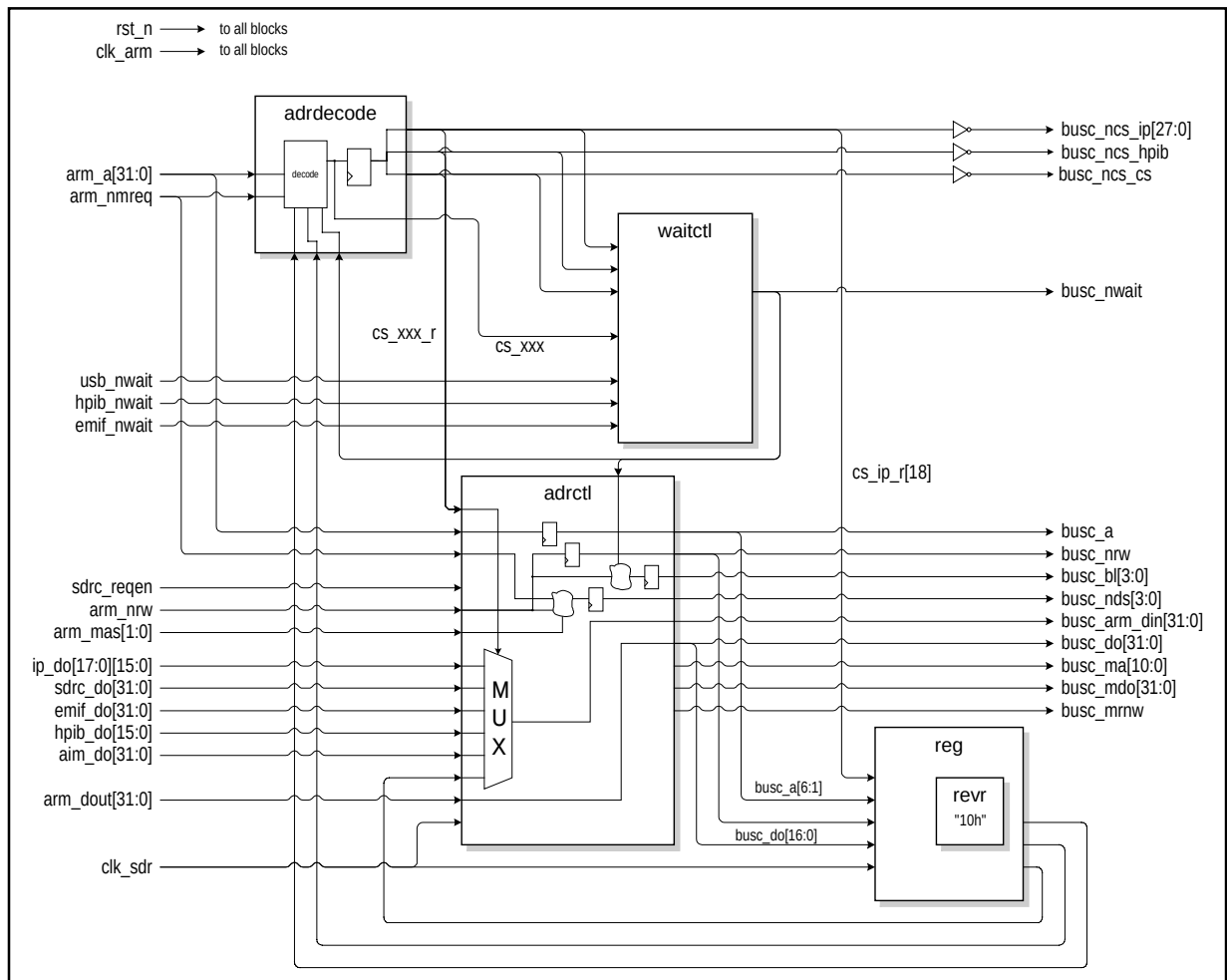


Figure 1 : Block Diagram of BUSC

4.2 Memory Map

ARM can access to all peripheral's registers, RAM and ROM area.

In addition, DSP memory can be also accessed by ARM through HPI function (Refer to HPI Bridge Specification).

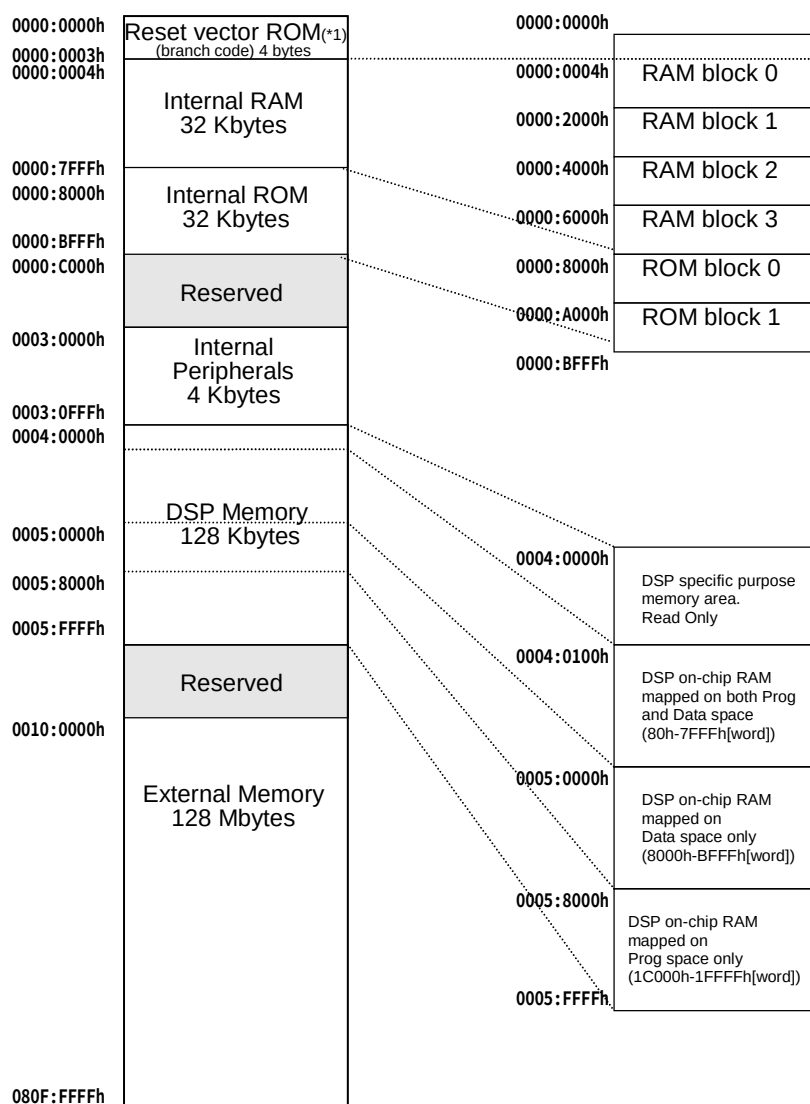
Boot ROM, SDRAM, expanded external memory and external Flash can be accessed through Memory Traffic Controller module. (Refer to Memory Traffic Controller Specification)

Internal RAM can be accessed directly by BUSC with 32 bits bus width.

The main decoding range are shown below table.

Table 1 : ARM Memory Space

Area Name	Start address	End address	Size in bytes
Reset vector ROM ^(*1)	0x0000:0000	0x0000:0003	4
ARM Internal RAM	0x0000:0004	0x0000:7FFF	32 K
ARM Internal ROM	0x0000:8000	0x0000:BFFF	16 K
Internal Peripheral	0x0003:0000	0x0003:0FFF	4 K
DSP Memory	0x0004:0000	0x0005:FFFF	128 K
External Memory	0x0010:0000	0x080F:FFFF	128 M



(*1): Reset vector ROM is used when BTSEL pin is set '0'. In this case, the first 4 byte of RAM block 0 (0000:0000h-0000:0004h) is not used. On the other hand, When BTSEL = '1', the area of Reset vector ROM is replaced by the first 4 byte of the RAM.

Figure 2 : ARM Memory Space Image

BUSC can be accessed to the internal peripheral modules with the data size of 16 bits.
The register address mapping is basically variable except USB, SD/MMC IF, MS, MG, UART and BUSC own registers.
The following table shows all the register's range to decode.

Table 2 : Internal Peripheral Memory Space

Peripheral Area Name	Mapping mode	Start address	End address	Size in half-word (16bits)
Peripheral 0	Variable	0x0003:0000	0x0003:007F	64
Peripheral 1	Variable	0x0003:0080	0x0003:00FF	64
Peripheral 2	Variable	0x0003:0100	0x0003:017F	64
Peripheral 3	Variable	0x0003:0180	0x0003:01FF	64
Peripheral 4	Variable	0x0003:0200	0x0003:027F	64
Peripheral 5	Variable	0x0003:0280	0x0003:02FF	64
Peripheral 6	Fixed	0x0003:0300	0x0003:037F	64
Peripheral 7	Fixed	0x0003:0380	0x0003:03FF	64
Peripheral 8	Variable	0x0003:0400	0x0003:047F	64
Peripheral 9 (SD/MMC IF)	Fixed	0x0003:0480	0x0003:04FF	64
Peripheral 10	Variable	0x0003:0500	0x0003:057F	64
Peripheral 11	Variable	0x0003:0580	0x0003:05FF	64
Peripheral 12	Variable	0x0003:0600	0x0003:067F	64
Peripheral 13	Variable	0x0003:0680	0x0003:06FF	64
Peripheral 14	Variable	0x0003:0700	0x0003:077F	64
Peripheral 15	Variable	0x0003:0780	0x0003:07FF	64
Peripheral 16	Variable	0x0003:0800	0x0003:087F	64
Peripheral 17	Variable	0x0003:0880	0x0003:08FF	64
Peripheral 18 (BUSC)	Fixed	0x0003:0900	0x0003:097F	64
Peripheral 19	Variable	0x0003:0980	0x0003:09FF	64
Peripheral 20	Variable	0x0003:0A00	0x0003:0A7F	64
Peripheral 21(USB)	Fixed	0x0003:0A80	0x0003:0AFF	64
Peripheral 22(USB)	Fixed	0x0003:0B00	0x0003:0B7F	64
Peripheral 23	Variable	0x0003:0B80	0x0003:0BFF	64
Peripheral 24(MG)	Fixed	0x0003:0C00	0x0003:0C7F	64
Peripheral 25(MS)	Fixed	0x0003:0C80	0x0003:0CFF	64
Peripheral 26	Variable	0x0003:0D00	0x0003:0D7F	64
Peripheral 27	Variable	0x0003:0D80	0x0003:0DFF	64

4.2.1. Reset Vector ROM

Reset Vector ROM (RVR), which is defined at 0x0000:0000 of ARM internal memory space (Refer to the table 1.), hard-coded in BUSC. ARM accessed this area to fetch instruction after power on reset. BUSC will provide ARM with the following code. The code means that jump to the address 0x0010:0000.

0000:0000 EA 03 FF FE ; RVR: B 0010_0000h

As the supplement this RVR code, the Branch and Branch with Link (B, BL) section of ARM datasheet is cited here.

Branch instructions contain a signed 2's complement 24 bit offset. This is shifted left two bits, sign extended to 32 bits, and added to the PC. The instruction can therefore specify a branch of +/- 32Mbytes. The branch offset must take account of the prefetch operation, which causes the PC to be 2 words (8 bytes) ahead of the current instruction.

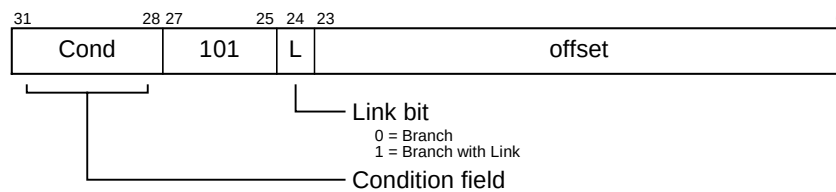


Figure 3 : Branch instructions

e.g.) Here are the example of how to calculate the offset for RVR code.

RVR: B #00100000h ; Jump to Flash memory with always condition.

00100000h - 8h = 000FFFF8h (subtract 2 words (8 bytes))
 000FFFF8h >> 2 = 3FFFEh (shift right two bits)

4.3 Chip Select

BUSC decodes the upper bits of the address bus and generates chip select signals (busc_ncs_ip), which is low active, to access the internal peripheral registers.

The chip select signals have to be connected to each peripheral module on several systems. Also the chip select signals for memory access are generated by BUSC.

The following table shows chip select signals and actual address decoding for DM270.

Table 3 : Chip Select Signal

Register / RAM / ROM	Peripheral/RAM/ROM Area Name	Chip Select	Start address	End address
Register	Timer 0	busc_ncs_ip[0]	0x0003:0000	0x0003:007F
	Timer 1	busc_ncs_ip[1]	0x0003:0080	0x0003:00FF
	Timer 2	busc_ncs_ip[2]	0x0003:0100	0x0003:017F
	Timer 3	busc_ncs_ip[3]	0x0003:0180	0x0003:01FF
	Serial 0	busc_ncs_ip[4]	0x0003:0200	0x0003:027F
	Serial 1	busc_ncs_ip[5]	0x0003:0280	0x0003:02FF
	UART 0	busc_ncs_ip[6]	0x0003:0300	0x0003:037F
	UART 1	busc_ncs_ip[7]	0x0003:0380	0x0003:03FF
	WDT	busc_ncs_ip[8]	0x0003:0400	0x0003:047F
	SD/MMC IF	busc_ncs_ip[9]	0x0003:0480	0x0003:04FF
	Interrupt Controller	busc_ncs_ip[10]	0x0003:0500	0x0003:057F
	GIO	busc_ncs_ip[11]	0x0003:0580	0x0003:05FF
	DSP Controller	busc_ncs_ip[12]	0x0003:0600	0x0003:067F
	OSD	busc_ncs_ip[13]	0x0003:0680	0x0003:06FF
	CCD Controller	busc_ncs_ip[14]	0x0003:0700	0x0003:077F
	Preview	busc_ncs_ip[15]	0x0003:0780	0x0003:07FF
	NTSC/PAL Encoder	busc_ncs_ip[16]	0x0003:0800	0x0003:087F
	Clock Controller	busc_ncs_ip[17]	0x0003:0880	0x0003:08FF
	Bus Controller	busc_ncs_ip[18]	0x0003:0900	0x0003:097F
	SDRAM Controller	busc_ncs_ip[19]	0x0003:0980	0x0003:09FF
	External Memory I/F (EM IF,CFC/SSF IF)	busc_ncs_ip[20]	0x0003:0A00	0x0003:0A7F
	USB0	busc_ncs_ip[21]	0x0003:0A80	0x0003:0AFF
	USB1	busc_ncs_ip[22]	0x0003:0B00	0x0003:0B7F
	3A	busc_ncs_ip[23]	0x0003:0B80	0x0003:0BFF
	MG	busc_ncs_ip[24]	0x0003:0C00	0x0003:0C7F
	Memory Stick	busc_ncs_ip[25]	0x0003:0C80	0x0003:0CFF
	ATM	busc_ncs_ip[26]	0x0003:0D00	0x0003:0D7F
	I2C	busc_ncs_ip[27]	0x0003:0D80	0x0003:0DFF
	ICACHE	busc_ncs_ip[28]	0x0003:0E00	0x0003:0E7F
Memory	DSP Memory (HPIB)	busc_ncs_hpib	0x0004:0000	0x0005:FFFF
	External Memory (SDRAM/Flash/CFC/SSF in same area)	busc_ncs_cs	0x0010:0000	0x080F:FFFF

The peripheral module, which has only registers to be accessed by ARM, needs to decode only the lower several bits of the address bus. The number of bit to be decoded the address bus in peripheral module is basically 6 bits (busc_a[6:1]). Then 6 bits of the address and the chip select can specify one register in peripheral from ARM.

4.4 Register Access

BUSC can access to all internal registers with 16 bits data width. A read and a write access can be achieved basically within one cycle of ARM clock except for USB, SD/MMC IF, MS and MG registers access (See the section of Handshaking Wait-state Control). The signal, which indicates to insert wait-state in read and write access and is synchronized on the rising edge of clk_arm clock, is required from peripheral module such like usb_nwait to access the registers from ARM.

The internal module should output all '0' on ARM data bus (ip_do) during inactivity of register selection. A half-word access is only allowed during register access from ARM as shown below table.

Table 4 : Cycle Number and Bus Width on Register access

Peripheral Module	Access Cycle Number	Bus Width [bit]	Access Mode
Timer 0	1	16	Half-word
Timer 1	1	16	Half-word
Timer 2	1	16	Half-word
Timer 3	1	16	Half-word
Serial 0	1	16	Half-word
Serial 1	1	16	Half-word
UART 0	2+n ^(*)	16	Half-word
UART 1	2+n ^(*)	16	Half-word
WDT	1	16	Half-word
SD/MMC IF	1+n ^(*)	16	Half-word
Interrupt Controller	1	16	Half-word
GIO	1	16	Half-word
DSP Controller	1	16	Half-word
OSD	1	16	Half-word
CCD Controller	1	16	Half-word
Preview	1	16	Half-word
NTSC/PAL Encoder	1	16	Half-word
Clock Controller	1	16	Half-word
Bus Controller	1	16	Half-word
SDRAM Controller	1	16	Half-word
External Memory I/F (EM IF, CFC/SSF IF)	1	16	Half-word
USB0	2+n ^(*)	16	Half-word
USB1	2+n ^(*)	16	Half-word
3A	1	16	Half-word
MG	2	16	Half-word
Memory Stick	1+n ^(*)	16	Half-word
ATM	1	16	Half-word
I2C	1	16	Half-word
ICACHE	1	16	Half-word

(*) n: This is variable number depending on access condition.

The following chart shows the register access to peripheral module.

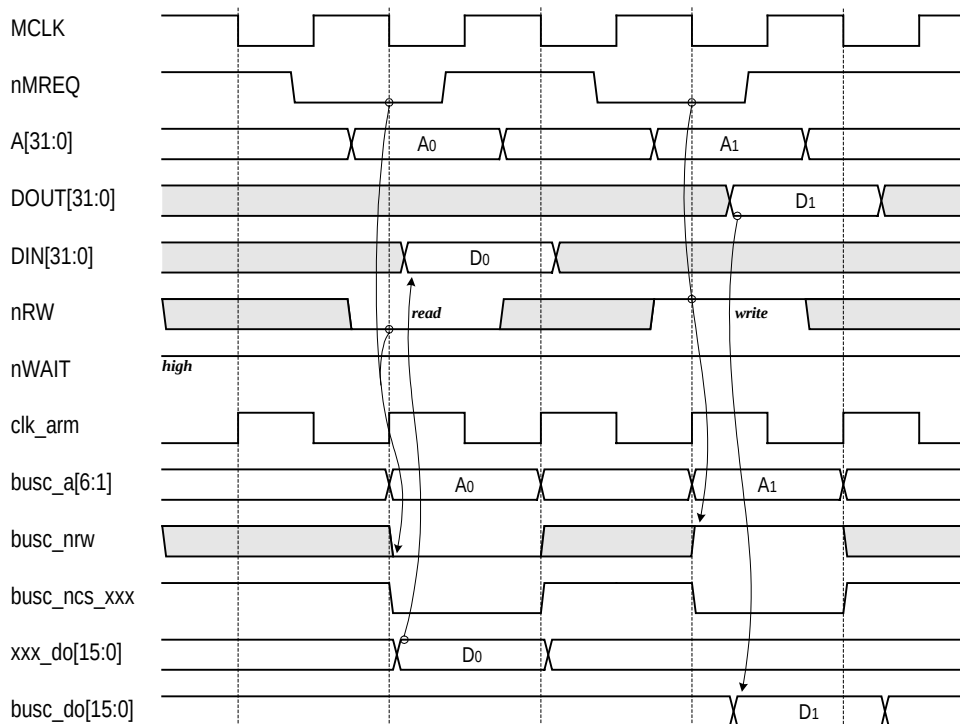


Figure 4 : Basic Peripheral's Register Access (no wait)

4.5 Access Mode and Data Strobe

There are a variety of transferring with some memories which has different bus width. The memory which is controlled by memory interface module can be accessed with a byte, a half-word and a word length. MAS[1:0] signal from ARM indicates the size of access to memory.

The data access mode are shown in the following table.

Table 5 : Data Access Mode

MAS[1:0]	Data Mode	Bit Width
00	Byte	8
01	Half-word	16
10	Word	32
11	reserved	-

During a half-word or a word access to peripheral, the address from ARM supposes being aligned. Hence, when ARM accesses to memory in a half-word or a word data mode, BUSC won't use A[0] and A[1:0] respectively during address decoding. The 16 bits peripheral doesn't look at lsb 0 of the internal address (busc_a[31:0]), the 32 bits peripheral doesn't look at busc_a[1:0].

For a byte (8 bits) or a half-word (16 bits) sized memory, BUSC generates Byte Latch (busc_bl[3:0]) signal and supplies ARM with data or instructions during read access. In a write access from ARM, BUSC generates data strobe signals (busc_nds[3:0]) for memory interface module to tell which data should be stored. This signal busc_nds[3:0] is active low. The following table shows data strobe signals and the variety of memory systems.

Table 6 : Data Strobe Generation

Peripheral	Data Mode	busc_a[1:0] decoded	busc_nds[3:0] (active low)	Target Memory
8 bits peripheral	Byte	00/01/10/11	- - - 0	Flash Memory / Optional External RAM SSF
	Half-word	00/01/10/11	- - - 0	Flash Memory
	Word		- - - 0	
16 bits peripheral	Byte	- 0	- - 1 0	Flash Memory
		- 1	- - 0 1	
	Half-word	0 - / 1 -	- - 0 0	Flash Memory / DSP Memory / CFC
	Word	0 - / 1 -	- - 0 0	Flash Memory
32 bits peripheral	Byte	00	1 1 1 0	SDRAM / External Memory
		01	1 1 0 1	
		10	1 0 1 1	
		11	0 1 1 1	
	Half-word	0 -	1 1 0 0	
		1 -	0 0 1 1	
	Word	- -	0 0 0 0	

 : not implemented for DM270

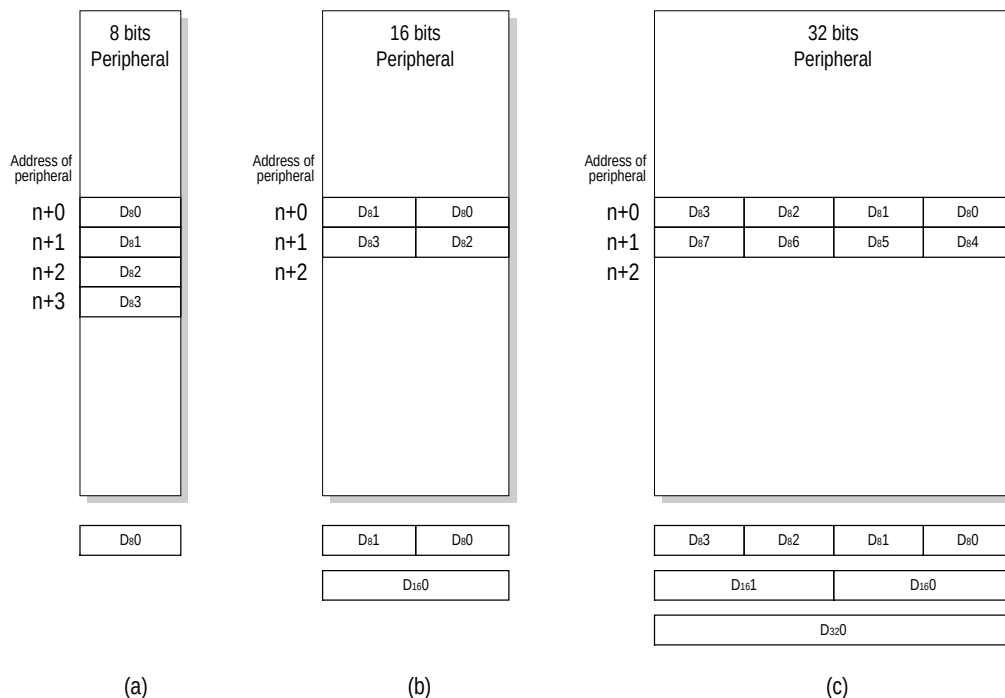


Figure 5 : Addressing image of 8/16/32 bits peripheral

The byte latch to be supplied to ARM required basically to be active the only corresponding bit which is requested data bus in the byte unit. But it's no problem that the byte latch signals are always active when ARM accesses to some data.

The following table shows the byte latch generation. Where in the items of busc_bl[3:0] (implemented), real values that BUSC module produced are shown.

Table 7 : Byte Latch Generation

Peripheral	Data Bus Width	Read / Write	Data Access Mode	busc_bl[3:0] (implemented)	busc_bl[3:0] (basically)
External Memory (Flash etc.)	32	R	Byte	1111	0001/0010/0100/1000
			Half-word	1111	0011/1100
			Word	1111	1111
		W	Byte	-	-
			Half-word		
			Word		
SDRAM	32	R	Byte	1111	0001/0010/0100/1000
			Half-word	1111	0011/1100
			Word	1111	1111
		W	Byte	-	-
			Half-word		
			Word		
ARM Internal Memory	32	R	Byte	1111	0001/0010/0100/1000
			Half-word	1111	0011/1100
			Word	1111	1111
		W	Byte	-	-
			Half-word		
			Word		
Peripheral Register	16	R	Byte	-	-
			Half-word	1111	0011
			Word	-	-
		W	Byte	-	-
			Half-word		
			Word		

4.6 Wait-state Generation

4.6.1 Fundamental Access

However it's better no insertion wait-state when ARM accesses to some memories, some wait cycles are required actually. In this section, fundamental access of a read or a write is described. In the case of access necessary additional wait-state, see the subsections of Handshaking Wait-state Control.

When ARM requests a half-word or a word data access to the some memories, wait-state is inserted automatically in the access cycle. The wait signal (busc_nwait) is asserted active low on the rising edge of clk_arm which has opposite phase to ARM core clock.

The following charts show the data or instruction read fundamental access with busc_bl[3:0] signals and wait-state insertion between BUSC and the peripheral which has different bus width.

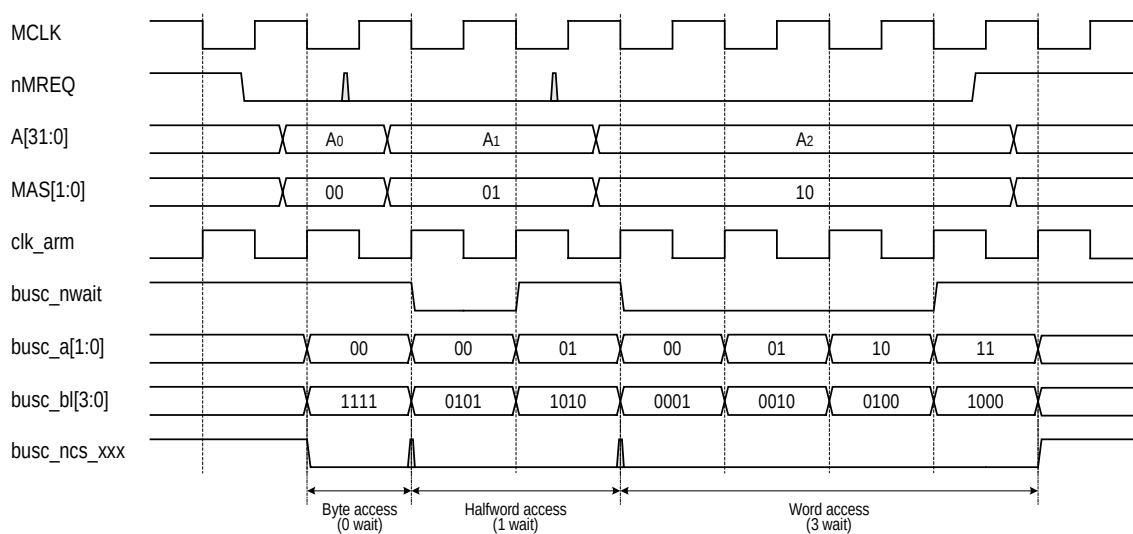


Figure 6 : Accessing 8 bits Peripheral with Byte Latch

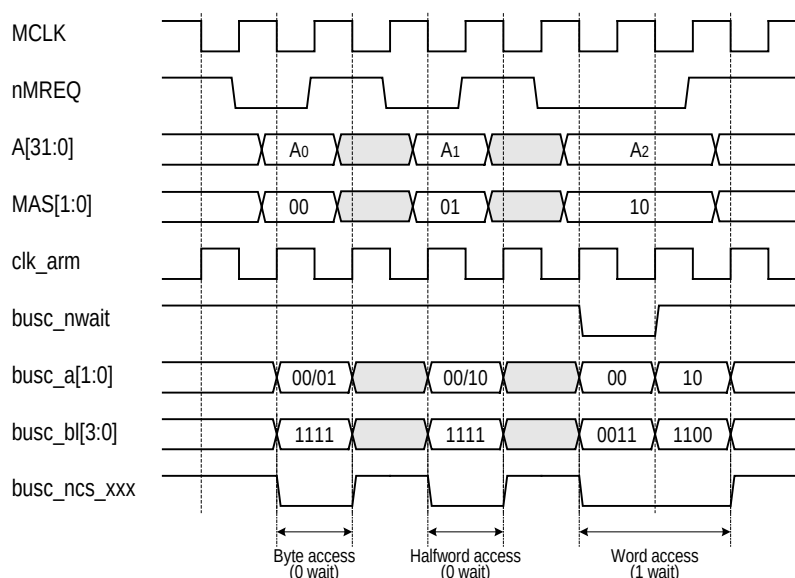


Figure 7 : Accessing 16 bits Peripheral with Byte Latch

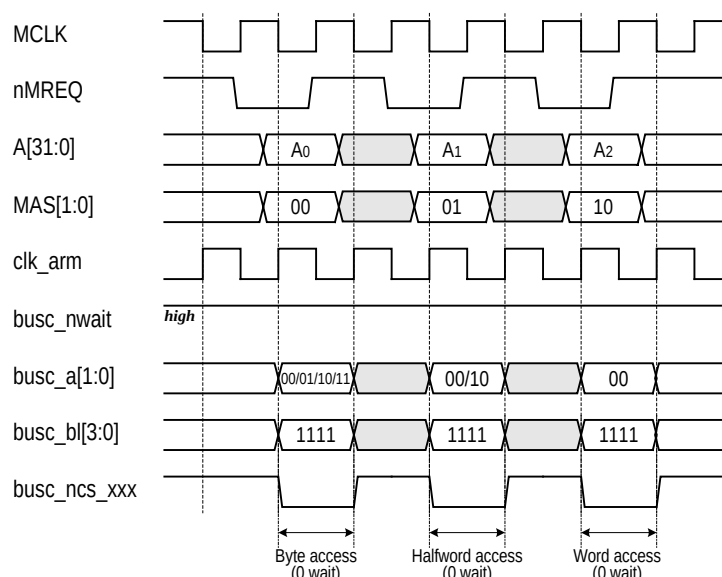


Figure 8 : Accessing 32 bits Peripheral with Byte Latch

4.6.2 Handshaking Wait-state Control

The some register and memory accesses cannot be achieved within one cycle of ARM clock. The busc_nwait may be stretched indefinitely by the wait-state request signal (xxxx_nwait) from peripheral to allow access to slow peripherals or memories.

In this subsection, the register and the memory handshaking access are described.

The requirement of handshaking access means that one wait-state is implicitly inserted automatically at the beginning of access.

The following table lists the peripherals and describes the handshaking wait-state control.

Table 8 : Wait-state Control

Peripheral	Handshake signal
Arm Internal Memory	aim_nwait
HPI Bridge	hpib_nwait
Memory Traffic Controller (via ICACHE)	lca_nwait_busc
USB(register access)	usb_nwait
SD/MMC IF	mmcif_nwait
Memory Stick	ms_nwait
UART0	uart0_nwait
UART1	uart1_nwait
Magic Gate	--- (fixed 2 cycle-width cs signal)

4.6.3 HPI Bridge Interface

When ARM accesses DSP Memory through HPI bridge module (HPIB), one wait-state is inserted automatically. In addition to it, extra wait cycles will be inserted depending on the wait signal which is generated in HPIB module.

The following chart shows basic access timing of HPIB module.

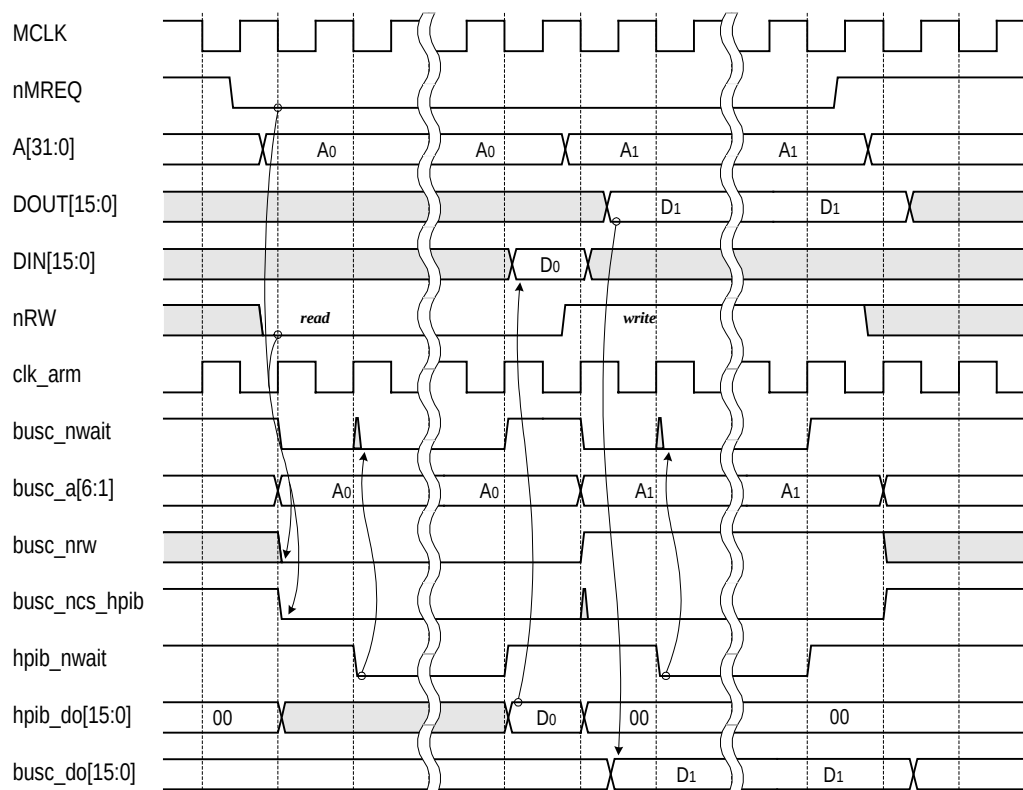


Figure 9 : HPI Bridge Access

4.6.4 Memory Traffic Controller Interface

External memory access timing is basically the same as HPI Bridge's. All signals between BUSC and MTC are synchronized with system clock (clk_arm). Refer to the document of External Memory Controller for details.

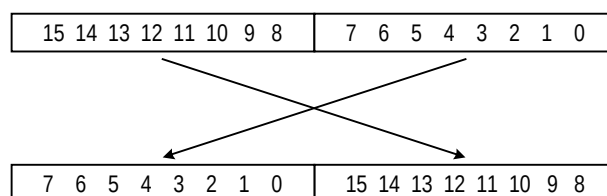
4.7 Endian Conversion

The endian can be converted by a use of Endian Convert Register (ECR).

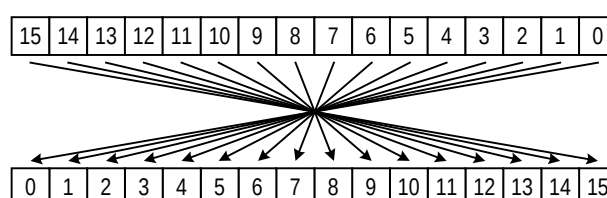
Writing data to be converted into ECR register once allows user to obtain either data which is converted by byte unit and by bit unit.

The converted data are stored to Endian Byte Converted Register (EBYTER) and Endian Bit Converted Register (EBITR).

The following is the image of endian conversion.



(a) Byte conversion



(b) Bit conversion

Figure 10 : Endian Conversion

4.8 External Memory

4.8.1 Chip Select

When ARM accesses the area of External Memory, BUSC outputs one chip-select signal (busc_ncs_cs) to Memory Traffic Controller module. The address range from 0x10:0000 to 0x80F:FFFF is assigned to External Memory area. In DM270, the address assignment for each external memory (SDRAM, flash, CFC etc.) is controlled by Memory Traffic Controller. Refer to the Memory Traffic Controller Specification for the address map.

5. REGISTERS

Table 9 : BUSC Register List

Register Name	Offset address	Description
ECR	0x00	Endian convert register
EBYTER	0x01	Endian byte converted register
EBITR	0x02	Endian bit converted register
REVR	0x03	Device revision register
AIMA	0x04	AIM Access control register, for key code input for reading CPRM ROM

ECR (0003:0900) Offset : 0x00

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	ECR15	ECR14	ECR13	ECR12	ECR11	ECR10	ECR09	ECR08	ECR07	ECR06	ECR05	ECR04	ECR03	ECR02	ECR01	ECR00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	ECR15-ECR00	The endian of written data is converted. The converted results can be read through EBYTER and EBITR registers.

EBYTER (0003:0902) Offset : 0x01

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	EBYT15	EBYT14	EBYT13	EBYT12	EBYT11	EBYT10	EBYT09	EBYT08	EBYT07	EBYT06	EBYT05	EBYT04	EBYT03	EBYT02	EBYT01	EBYT00
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	EBYT15-EBYT00	The data which is swapped the upper byte for the lower byte can be read when the data to be converted the endian is written to ECR register.

EBITR (0003:0904) Offset : 0x02

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	EBIT15	EBIT14	EBIT13	EBIT12	EBIT11	EBIT10	EBIT09	EBIT08	EBIT07	EBIT06	EBIT05	EBIT04	EBIT03	EBIT02	EBIT01	EBIT00
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	EBIT15-EBIT00	The data which is swapped each bit can be read when the data to be converted the endian is written to ECR register.

REVR (0003:0906) Offset : 0x03

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	REV7	REV6	REV5	REV4	REV3	REV2	REV1	REV0
R/W	-	-	-	-	-	-	-	-	R	R	R	R	R	R	R	R
Default	-	-	-	-	-	-	-	-	0	0	0	1	0	0	0	0

Bits	Register Name	Description
15:8	RSV	Reserved bit.
7:0	REV7-REV0	Device revision code. eg.) ES1.0 => 0x10

AIMA (0003:0908) Offset : 0x04

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	AIMA15	AIMA14	AIMA13	AIMA12	AIMA11	AIMA10	AIMA09	AIMA08	AIMA07	AIMA06	AIMA05	AIMA04	AIMA03	AIMA02	AIMA01	AIMA00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	AIMA15-AIMA00	Register for key code enables ARM to access CPRM ROM data

6. Interface

Table 10 : BUSC Top-level I/O List

Pin Name	Bits	I/O	Fr/To	Description
Functional clocking / common signals				
rst_n	1	I	ext	Power on reset..
clk_arm	1	I	clkc	ARM clock.
busc_a	32	O	(*1)	Address bus.
busc_do	32	O	(*1)	Data output bus.
busc_nds	4	O	(*1)	Data strobe. Active low.
ARM interface signals				
arm_nmreq	1	I	ARM	Memory request from ARM.
arm_a	32	I	ARM	Address bus from ARM.
arm_dout	32	I	ARM	Data output bus from ARM.
busc_arm_din	32	O	ARM	Data input bus to ARM.
arm_nrw	1	I	ARM	Read/write direction from ARM. 0:Read / 1:write.
arm_mas	2	I	ARM	Memory access size from ARM.
arm_nopc	1	I	ARM	'not op-code fetch' signal from ARM
busc_nwait	1	O	ARM	Active low wait to ARM.
busc_bl	4	O	ARM	Byte latch to ARM.
Peripheral module interface signals				
busc_nrw	1	O	(*1)	0:Read / 1:write. Active low.
busc_ncs_ip	29	O	(*1)	chip select. Active low.
busc_ncs_hpib	1	O	hpib	HPI bridge chip select. Active low.
busc_ncs_cs	1	O	ica	MTC chip select via ICACHE. Active low.
busc_nopc	1	O	ica	'not op-code fetch' signal to ICACHE
ip_do[28:0]	16	I	(*1)	Data input bus.
ica_do	32	I	ica	Memory Traffic Controller data input bus. (via ICACHE)
hpib_do	16	I	hpib	HPI bridge data input bus.
Internal memory interface signals				
aim_do	32	I	aim	Internal memory data input bus.
aim_do_cprm	32	I	aim	ROM secure code read bus
aim_nwait	1	I	aim	Wait signal to access AIM. Active low.
busc_ma	11	O	aim	Internal memory address bus.
busc_mdo	32	O	aim	Internal memory data output bus.
busc_ncs_aim	6	O	aim	Internal memory chip select. Active low.
busc_cprm_req	1	O	aim	CPRM secure code download request signal
busc_rom_readen	1	O	aim	CPRM ROM area access enable signal
busc_mnrw	1	O	aim	Internal memory 0:Read / 1:Write.
HPIB access interface signals				
hpib_nwait	1	I	hpib	Wait signal to access HPI. Active low.
Memory Traffic Controller signals via ICACHE				
ica_nwait	1	I	ica	Wait signal to access external memory. Active low.
USB interface signal				
usb_nwait	1	I	usb	Wait signal to access registers of USB. Active low.
SD/MMC interface signal				
mmcif_nwait	1	I	mmcif	Wait signal to access registers of SD/MMC IF. Active low.
UART interface signal				
uart0_nwait	1	I	uart0	Wait signal to access registers of UART0. Active low.
uart1_nwait	1	I	uart1	Wait signal to access registers of UART1. Active low.
Memory Stick interface signal				
ms_nwait	1	I	ms	Wait signal to access registers of MS IF. Active low.
CLKC interface signal				

Pin Name	Bits	I/O	Fr/To	Description
clk_en_rom	1	I	clk	Clock enable signal for CPRM ROM
Upper level module				
btsel	1	I	upper	Boot select signal
revr	8	I	upper	Device revision
Test				
tst_scan	1	I	tst	Scan test enable

(*1) : Internal peripheral modules

7. TEST Methodology

None.